Data Quality Checklist for New Pipelines

Use this checklist before a new pipeline goes to production. Every item should be checked by the developer and reviewed by a second engineer.

1. Completeness

Compare row counts between source and target after each load. A difference of more than 1 percent must raise an alert. Check that mandatory columns contain no null values.

2. Uniqueness

Every table must have a primary key. Add a dbt unique test and a not_null test on the primary key column of every staging and mart model.

3. Freshness

Configure dbt source freshness with warn_after of 12 hours and error_after of 24 hours for daily sources. Freshness checks should run before the main dbt build.

4. Validity

Use accepted_values tests for status columns (for example order status must be one of placed, shipped, completed, returned). Use relationships tests to make sure foreign keys exist in the parent table.

5. Monitoring and ownership

Each pipeline must have a named owner, an alert channel, and a runbook link. Failed data quality checks block downstream models from running.

6. Sign-off

The pipeline can be promoted to production only when all checks pass in the staging environment for three consecutive days.